The Power Rule (template)

You missed this lesson. Copy this into your notes. That is all you need to do.

Question of the day

How fast does x? change, for any n?

Targets

- I can state the power rule in words.
- I can differentiate a polynomial term by term.
- I can find the slope at a point.

Concepts

- **Derivative** (f?(x)): The slope of the curve at a point. The rate of change of the output.
- **Power rule** (d/dx x? = n x?$^{-1}$): Bring the power down in front. Then lower the power by 1.

What the class predicted, and what is true

- The derivative of x^2 is $2x$. What do you expect for x^3? $3x^2$. Bring the 3 down and lower the power: $3x^2$.
- What is the slope of the flat line $y = 7$? **0.** A flat line does not rise: the derivative is 0.

Huddle notes (answers filled in)

Definitions

- **Derivative:** The **slope** of the tangent line at a point.
- **Power rule:** Bring the **power** down, then **subtract 1** from it.

Worked example

Differentiate f(x) = 4x^3.

- Bring down the power: 3 · 4x
- Lower the power by 1: 12x**2**
- So f?(x) = **12x^2**

Label the slope

1. The **point** where we measure the slope.
2. The **tangent line**; its slope is f?.

One worked sample of each skill

Differentiate

Level 1. Differentiate.
Answer: $4x^3$ Bring down 4 and lower the power: $4x^3$.

Level 2. Differentiate.
Answer: $12x^1$ Multiply 6 by 2 to get 12, then lower the power: $12x^1$.

Level 3. Differentiate term by term.
Answer: $10x^4 + 5$ $10x^4 + 5 + 0$. The constant 1 disappears.

Level 4. Differentiate. Rewrite the root as a power if that helps.
Answer: $\frac{6}{2\sqrt{x}}$ $6x^{1/2}$ becomes $\frac{6}{2}x^{-1/2} = \dfrac{6}{2\sqrt{x}}$.

Slope at a point

Level 1. Let $f(x) = x^2$. Find the slope of the curve at $x = 2$.
Answer: 4 $f'(x) = 2x^1$, so $f'(2) = 4$.

Level 2. Let $f(x) = 4x^2$. Find the slope of the curve at $x = 3$.
Answer: 24 $f'(x) = 8x^1$, so $f'(3) = 24$.

Level 3. Let $f(x) = 2x^4$. Find the slope of the curve at $x = 3$.
Answer: 216 $f'(x) = 8x^3$, so $f'(3) = 216$.

Level 4. Let $f(x) = 4x^4$. Find the slope of the curve at $x = 1$.
Answer: 16 $f'(x) = 16x^3$, so $f'(1) = 16$.